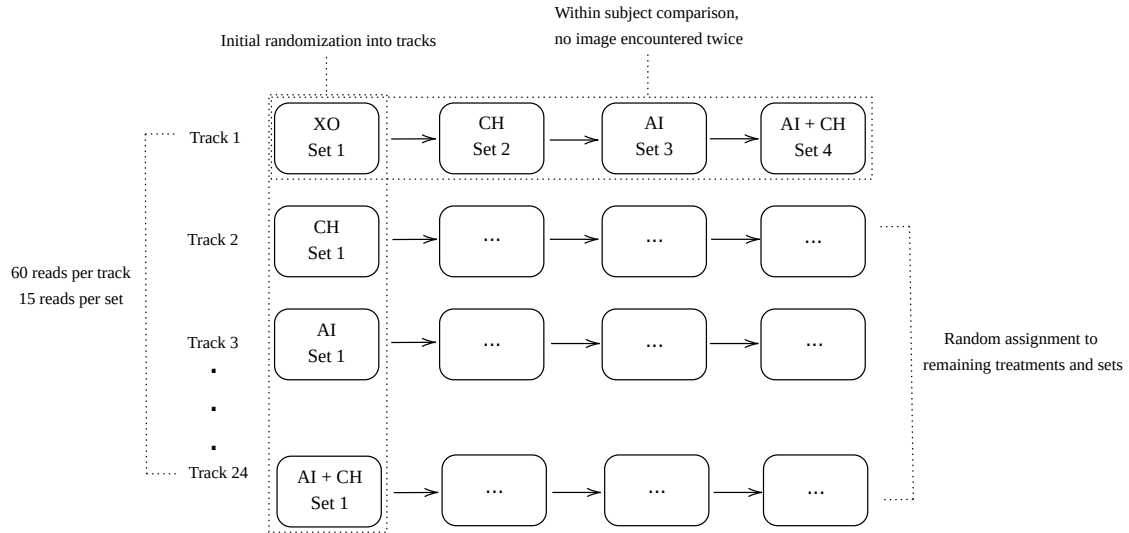


B Appendix of Experimental Interface and Instructions

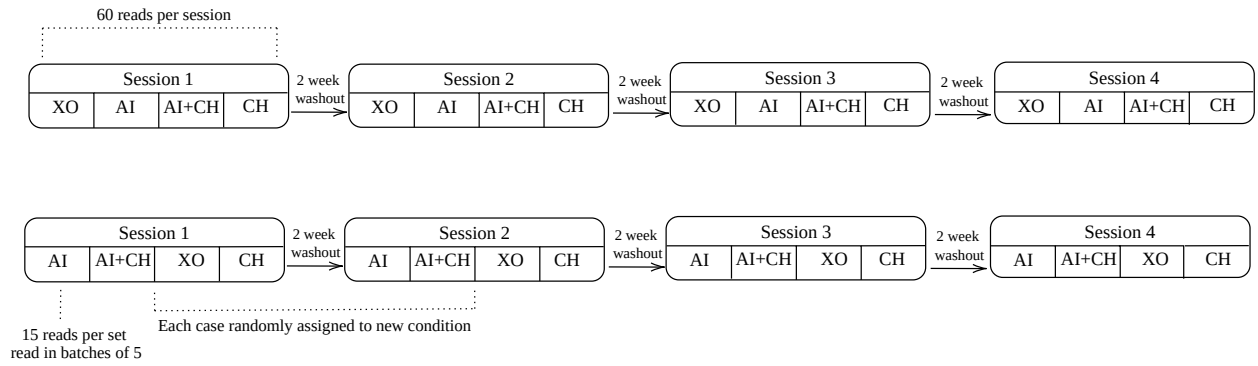
B.1 Design

Figure B.1: Design 1



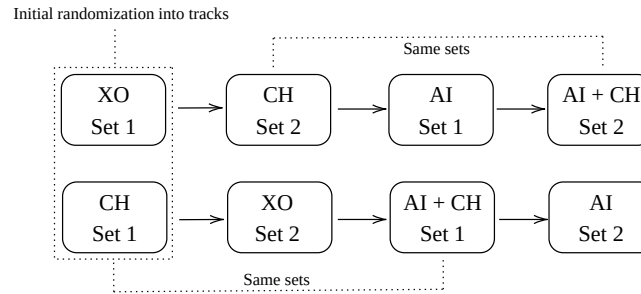
Note: In this design, radiologists are assigned to a randomized sequence of the four information environments., resulting in twenty-four possible tracks. Under each information environment they read fifteen cases. Radiologists encounter each patient case at most once. At the beginning of the experiment every radiologist reads eight practice cases. Furthermore, a random half of the participating radiologists receive incentives for accuracy.

Figure B.2: Design 2



Note: In this design, radiologists diagnose sixty patient cases each under the four information environments. Radiologists read every case under every information environment across four sessions, separated by a washout period. Each case is only encountered once per session and to ensure that radiologists do not recall their/AI predictions from previous reads of the same cases, we ensure a minimum two-week washout period between two subsequent sessions. Within every experimental session they therefore read fifteen under each information environment. The randomization occurs at the track-level where every track has a different sequence of the information environments. (Example tracks shown here.)

Figure B.3: Design 3



Note: In this design, radiologists diagnose fifty cases, first without and then with AI assistance. Clinical history is randomly provided in either the first or second half of images forming the basis of the randomization. The cases diagnosed with and without clinical history are different.

B.2 Instructions

Below you will find the instructions received by the subjects when receiving the interface-based treatment. Comments on the instructions are provided in italics and were not seen by subjects.

Instructions

You are about to participate in a study on medical decision making. You may pause the study at any time. To resume, revisit the link you were given and your progress will have been saved.

We will present you with adult patients with potential thoracic pathologies. These patients will be presented under the following four scenarios:

1. Only a chest X-ray is shown.
2. An X-ray is accompanied with additional information about the clinical history.
3. An X-ray is shown along with Artificial Intelligence (AI) support. This AI tool is described in further detail below.
4. An X-ray is shown along with both additional information on clinical history and the AI support.

The patients are randomly assigned to each of these scenarios. That is, availability of clinical history and/or AI support is unrelated to the patient.

Clinical History: includes available lab results or indications by the treating physician, if any.

AI support: This tool uses only the X-ray image to predict the probability of each potential pathology of interest. The tool is based on state-of-the-art machine learning algorithms developed by a leading team of researchers at Stanford University.

Responses

For each patient and pathology, we will ask for both an assessment and a treatment decision:

1. We will first ask for your assessment of the probability that each condition is present in a patient. **Please consider all pathologies and findings that would be relevant in a radiology report for the patient. You should express your uncertainty about the presence of one or many conditions by appropriately choosing the probability.** Note that it is possible that the patient has multiple such conditions or none of them.
2. If you determine that a pathology may be present, we may ask you to rate the severity and/or extent of the disease on a scale.
3. Finally, when relevant we will ask whether you would recommend treatment or follow up according to the clinical standard of care if you determine that the pathology may be present. The first two responses are diagnostic while the third is a clinical decision. We are aware that a single physician or radiologist typically does not perform both tasks. However, for this study, we ask that you respond to the best of your ability in both of these roles.

Browser Compatibility

This platform supports desktop versions of Chrome, Firefox, and Edge. Important features on non-supported browsers (including Safari) are missing and we discourage their use for this experiment. In addition, the platform does not support any mobile devices and the platform will perform poorly on mobile. If you encounter any issues during the experiment, please send an email to DiagnosticAI@mit.edu and we will follow-up quickly.

Hierarchy

The interface uses a hierarchy to categorize various thoracic conditions. It will be useful to familiarize yourself with this hierarchy before you start, but you may also revisit the hierarchy at any time throughout the experiment by clicking the help tab in the upper right corner.